

Paper Review: Smoking Bans, Maternal Smoking and Birth Outcomes

ECMT677 Applied Microeconometrics
Texas A&M University

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Summary of the paper

1. Summary of the paper

- Motivation & Questions
- Identification Strategy
- Channels Considered
- Results
- Robustness Check

2. Critics of the paper

- Common Trend Assumption
- Effect of Substitution Behavior
- Effect of Job Switching Status

Motivation & Questions

Motivation:

- One of the negative externality of smoking is that it might do harm to non smokers.
- Norwegian Smoking ban in 2004 provides a good source of variation to study this externality.
 - In Norway, smoking has been prohibited in all public places except the hospitality industry (e.g. bars, restaurants).
 - In 2004, the ban was extended to this industry.

Research Questions:

- **How does smoking ban affect birth outcome of babies born to females working in bars or restaurants?**
- **Do changes in second hand exposure for non smoking females who work in bar or restaurants have effect on birth outcome?**

Identification Strategy: Difference in Difference

- Compare mothers working in bars and restaurants (the treatment group) to mothers in stores (the control group)
 - **Source of variation:** Extension of smoking ban to bars and restaurants on June 1st, 2004 in Norway
- **Assumption:**

The expected change over time for the birth outcomes from the mothers working in a similar occupation (stores) provides a good **counterfactual** for the expected change for the birth outcomes from the mothers working in bars and restaurants in the absence of treatment.

- **Regression equation:**

$$Y_{ijt} = \alpha_1 + \alpha_2 Treat_j + \alpha_3 Treat_t + \alpha_4 (Treat_j * Post_t) + \alpha_5 X_{ijt} + \varepsilon_{ijt}$$

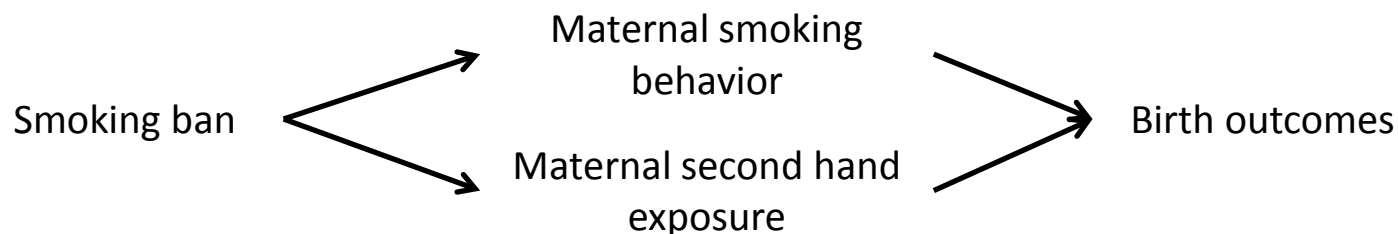
where i indexes individual, j the workplace, and t time.

Treat = 1 if mothers work in bars or restaurants as of 2003

Post = 1 if t is after June 1st 2004

Channels Considered

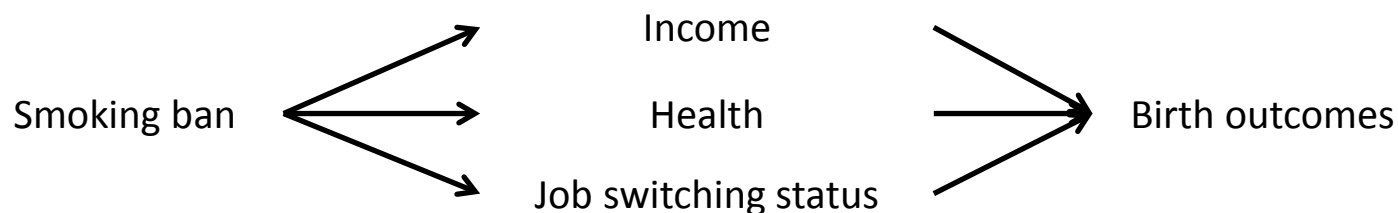
- Main Channels (at workplace):**



- Effect of Substitution Behavior (at home):**



- Additional Channels Considered:**



Results: Effect of Smoking Ban on Birth Outcomes

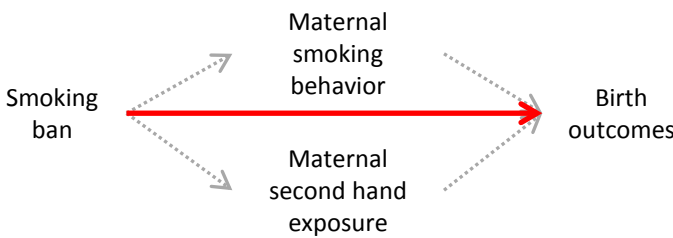


Table 1. The effect of smoking ban on birth outcomes

	Birth weight	<1000g	<1500g	<2000g	<2500
Post *Treated	57.69 (1.28)	-.0033 (-.46)	-.0189** (-2.51)	-.0254** (-2.75)	-.0006 (-.05)
Post	-13.66 (-.66)	.0010 (.25)	.007 (1.04)	.0058 (1.14)	-.0060 (-.83)
Treated	-54.05** (-2.21)	.0042 (1.13)	.012*** (2.95)	.012** (2.52)	-.0003 (-.03)
N	4007	4007	4007	4007	4007
	Pre-term birth	APGAR 1 min	APGAR 5 min	Boy	Birth defect
Post *Treated	-.0255* (-1.79)	-.0283 (-.32)	-.0494 (-.58)	.0388 (.40)	-.0003 (-.02)
Post	.0016 (.24)	.0093 (.16)	-.0263 (-.51)	.0044 (1.22)	-.0135 (-1.32)
Treated	.0115 (1.44)	-.0161 (-.30)	-.0244 (-.48)	.00246 (.57)	-.008 (-.48)
N	4004	4000	4000	4007	4007

Notes: We use a 5 month window, we cluster on treated*birth month, controls included in regressions are mother’s income in 2003; age of mother at birth; mother’s years of education; mother’s work hours in 2003; dummies for birth order; single birth; and dummies for each county of birth (19 counties). *p<0.10, **p<0.05, ***p<0.01.

Significant effect for
VLBW(very low birth weight)
and pre-term birth

Results: Effect of Smoking Ban on Mother’s Smoking Status

- 3rd Column shows that differential effect of smoking ban on quitting smoking by the end of the pregnancy for the subset of mothers that reported to smoke at the start of the pregnancy.
 - Mothers in treatment group are 15% points more likely to quit

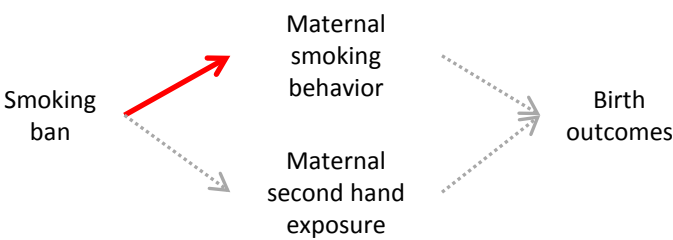


Table 2. The effect of smoking ban on mother’s smoking status

	Mother smoke start	Mother no info	Quit smoking
Post *Treated	-.0309 (-1.05)	.0221 (.66)	.154** (2.15)
Post	.0228 (1.45)	.0214 (1.46)	-.0758** (-2.27)
Treated	.0660*** (3.57)	-.0019 (-.15)	-.132** (-2.42)
N	3004	3532	767

Notes: We use a 5 month window, we cluster on treated*birth month, controls included in regressions are mother’s income in 2003; age of mother at birth; mother’s years of education; mother’s work hours in 2003; dummies for birth order; single birth; and dummies for each county of birth (19 counties). *p<0.10, **p<0.05, ***p<0.01.

Results: Effect of Smoking Ban on Birth Outcomes by Smoking Status

- 2nd row shows that differential effect of smoking ban on birth outcome for non smoking mothers
 - can be interpreted the effect through 2nd hand smoking channel
 - effects are smaller than in smoking mothers and insignificant

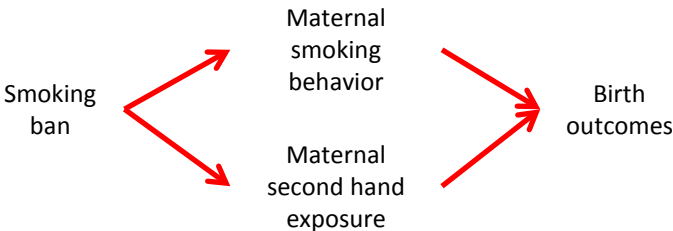


Table 3. The effect of smoking ban on birth outcomes: by smoking status

a) Mother smokes at start of pregnancy	Birth weight	<1000g	<1500g	<2000g	<2500	Pre-term birth	APGAR 1 min	APGAR 5 min	Boy	Birth defect
Post *Treated	162.5**	-.0107	-.0290*	-.0290	.0065	-.0298	.0580	.0069	-.0060	-.0261
	(2.17)	(-1.18)	(-1.87)	(-1.16)	(.20)	(-.90)	(.21)	(.03)	(-.09)	(-.98)
N	793	793	793	793	793	793	793	793	793	793
b) Mother does not smoke at start of pregnancy	Birth weight	<1000g	<1500g	<2000g	<2500	Pre-term birth	APGAR 1 min	APGAR 5 min	Boy	Birth defect
Post *Treated	14.70	.00222	-.0058	-.0212	-.0025	-.0154	-.0103	-.0178	.0131	.0062
	(.24)	(.20)	(-.53)	(-1.68)	(-.16)	(-.55)	(-.09)	(-.16)	(.32)	(.27)
N	2548	2548	2548	2548	2548	2548	2548	2548	2548	2548
c) Missing information on mother's smoking status	Birth weight	<1000g	<1500g	<2000g	<2500	Pre-term birth	APGAR 1 min	APGAR 5 min	Boy	Birth defect
Post *Treated	105.5	-.0144	-.0477**	-.0323	.0099	-.0642	-.0818	-.114	.0660	.016
	(1.27)	(-.90)	(-2.73)	(-1.16)	(.28)	(-1.64)	(-.33)	(-.45)	(.67)	(.45)
N	642	642	642	642	642	642	642	642	642	642

Notes: We use a 5 month window, we cluster on treated*birth month, controls included in regressions are mother's income in 2003; age of mother at birth; mother's years of education; mother's work hours in 2003; dummies for birth order; single birth; and dummies for each county of birth (19 counties). *p<0.10, **p<0.05, ***p<0.01.

Robustness Check: Placebo test

- Test if there is any effect of fake treatment (pretending that smoking ban is in June 2002)
 - To see if the results are driven by smoking ban rather than any other event or seasonality

Table 8. Placebo effects of smoking ban on birth outcomes (pretend reform is in June 2002)

	Birth weight	<1000g	<1500g	<2000g	<2500
Post *Treated	25.66 (.81)	-.0011 (-.21)	.0041 (.67)	.0055 (.70)	-.0000 (-.00)
Post	33.60* (1.74)	-.0022 (-.61)	-.0042 (-.96)	-.0098** (-2.21)	-.0114* (-1.78)
Treated	-20.37 (-.76)	-.0009 (-.27)	-.0038 (-1.23)	.0008 (.13)	.0037 (.37)
N	3710	3710	3710	3710	3710
	Pre-term birth	APGAR 1 min	APGAR 5 min	Boy	Birth defect
Post *Treated	-.0097 (-.85)	-.0333 (-.56)	.0616 (1.07)	.0116 (.27)	-.0162 (-1.16)
Post	-.0110 (-1.71)	.0035 (.13)	-.0109 (-.51)	.0026 (.19)	.0179** (2.20)
Treated	-.0004 (-.06)	.0598* (1.77)	-.0048 (-.09)	-.0114 (-.76)	.0066 (.67)
N	3689	3689	3710	3710	3710

Notes: We use a 5 month window, we cluster on treated*birth month, controls included in regressions are mother’s income in 2001; age of mother at birth; mother’s years of education; mother’s work hours in 2001; dummies for birth order, single birth; and dummies for each county of birth (19 counties). *p<0.10, **p<0.05, ***p<0.01.

Summary of the paper

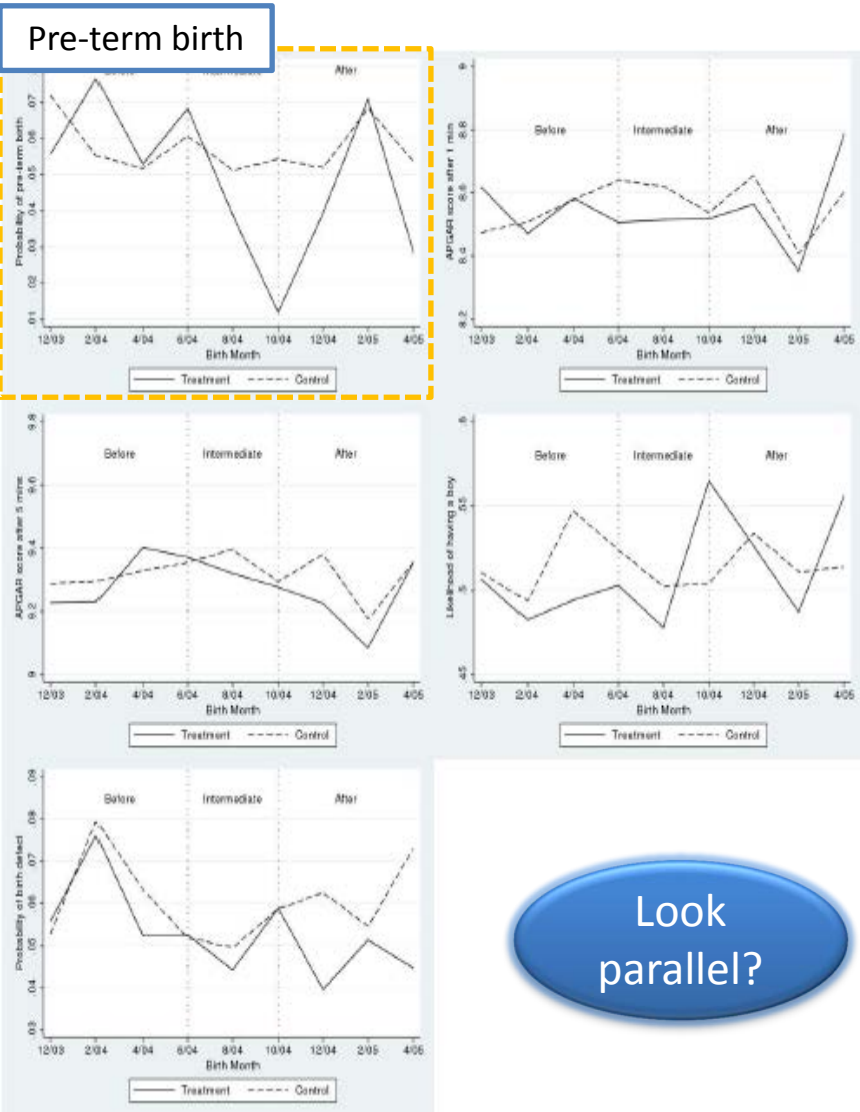
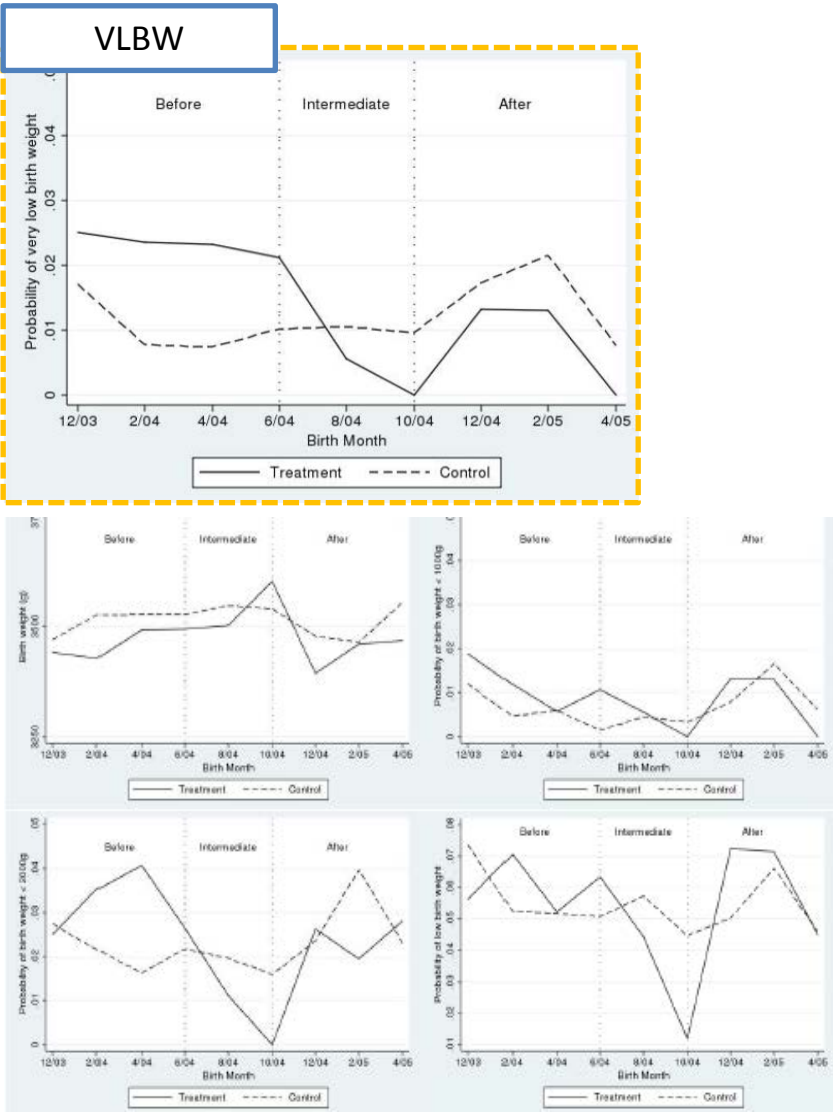
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Common Trend Assumption



Look parallel?

Effect of Substitution Behavior

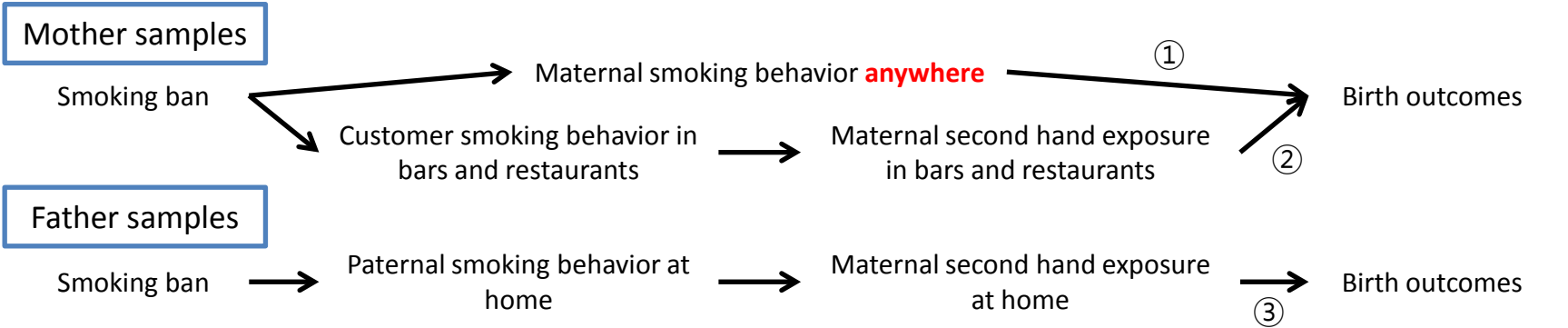


Table 7. Fathers: The effect of smoking ban on birth outcomes

	Birth weight	<1000g	<1500g	<2000g	<2500
Post *Treated	-77.09* (-1.90)	-.0067 (-1.42)	-.0036 (-.51)	.0051 (.52)	.0028 (.23)
Post	24.84 (.88)	.0022 (1.00)	.00322 (1.02)	-.034 (-.68)	-.0084 (-.94)
Treated	-25.63 (-1.00)	.0041 (1.12)	.0056 (1.04)	-.0022 (-.31)	-.0041 (-.43)
N	3941	3941	3941	3941	3941

	Pre-term birth	APGAR 1 min	APGAR 5 min	Boy	Birth defect
Post *Treated	.0114 (.97)	-.101 (-1.33)	.0492 (1.05)	-.0159 (-.58)	.0093 (.70)
Post	-.0064 (-.63)	.123*** (4.01)	.0401* (1.86)	-.0102 (-.64)	-.0055 (-.68)
Treated	-.0159* (-1.96)	.0206 (.34)	-.0244 (-.60)	.0264 (1.29)	-.0272*** (-2.75)
N	3915	3933	3933	3941	3941

Notes: We use a 9 month window, we cluster on treated*birth month, controls included in regressions are mother's income in 2003; age of mother at birth; mother's years of education; mother's work hours in 2003; dummies for birth order; single birth; and dummies for each county of birth (19 counties). *p<0.10, **p<0.05, ***p<0.01.

- Worse outcomes?
- Mothers are more likely to quit smoking (including at home), but fathers are likely to smoke more at home (with smoking status unknown)?
- For the fathers, is the effect of smoking ban on substitution behavior stronger than the effect on quitting smoking?

Effect of Job Switching Status

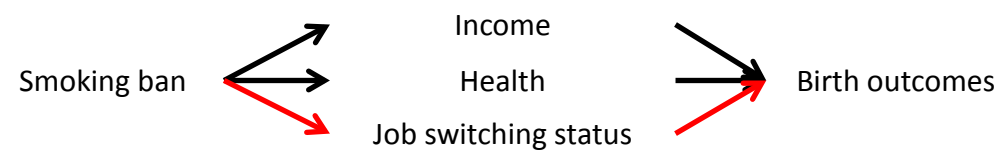


Table 4. The effect of smoking ban on mother’s outcomes

	Log Income 2004	Log Income 2005	Sick at least 1 out of last 3 months of pregnancy	Sick all of last 3 months of pregnancy
Post *Treated	.0644 (1.02)	-.0283 (-.34)	-.0120 (-.07)	.0008 (.03)
Post	.130*** (7.85)	.309*** (13.91)	-.121 (-1.01)	-.0128 (-.96)
Treated	-.0824** (-2.36)	-.0104 (-.14)	.0298 (.33)	.0131 (.91)
N	3972	3908	4007	4007

	Switch job 2004	Switch job 2005
Post *Treated	.043* (2.02)	.0844* (1.87)
Post	.0613*** (3.89)	-.0174 (-1.07)
Treated	.0347* (1.99)	.0861** (2.72)
N	2917	2668

Notes: We use a 5 month window, we cluster on treated*birth month, controls included in regressions are mother’s income in 2003; age of mother at birth; mother’s years of education; mother’s work hours in 2003; dummies for birth order; single birth; and dummies for each county of birth (19 counties). *p<0.10, **p<0.05, ***p<0.01.

Table 5. The effect of smoking ban on birth outcomes for non-switchers

	Birth weight	<1000g	<1500g	<2000g	<2500
Post *Treated	71.78 (1.02)	-.0082 (-.84)	-.0208* (-1.88)	-.0275** (-2.45)	.0102 (.51)
N	2625	2625	2625	2625	2625

	Pre-term birth	APGAR 1 min	APGAR 5 min	Boy	Birth defect
Post *Treated	-.005 (-.20)	-.0110 (-.08)	-.0178 (-.14)	.0719* (1.94)	.0094 (.38)
N	2625	2621	2622	2625	2625

Still valid (?) but why no result for switchers?

Table 6. Compositional changes of switchers

Mother	Age at birth	Wage in 2003	Years of schooling	Smokes	Does not smoke	No info on smoke
Post *Treated*Switching	1.854 (1.38)	25392 (.93)	.0776 (.08)	-.0317 (-.24)	.0488 (.52)	.001 (.01)
N	2917	2917	2917	2917	2917	2917

Notes: We use a 5 month window, we cluster on treated*birth month, controls included in regressions are mother’s income in 2003; age of mother at birth; mother’s years of education; mother’s work hours in 2003; dummies for birth order; single birth; and dummies for each county of birth (19 counties). *p<0.10, **p<0.05, ***p<0.01.

Why no result on quitting smoking?